

Pencitraan digital 2

Image enhancement using histogram equalization

Aim and method. Improve a low-contrast grayscale image and explain the result using before-and-after histograms. The supplied input is a deliberately flattened version of a NASA sample photograph. It is loaded with OpenCV and processed with `cv2.equalizeHist(gray)`. No mathematical derivation or reference-quality score is used.

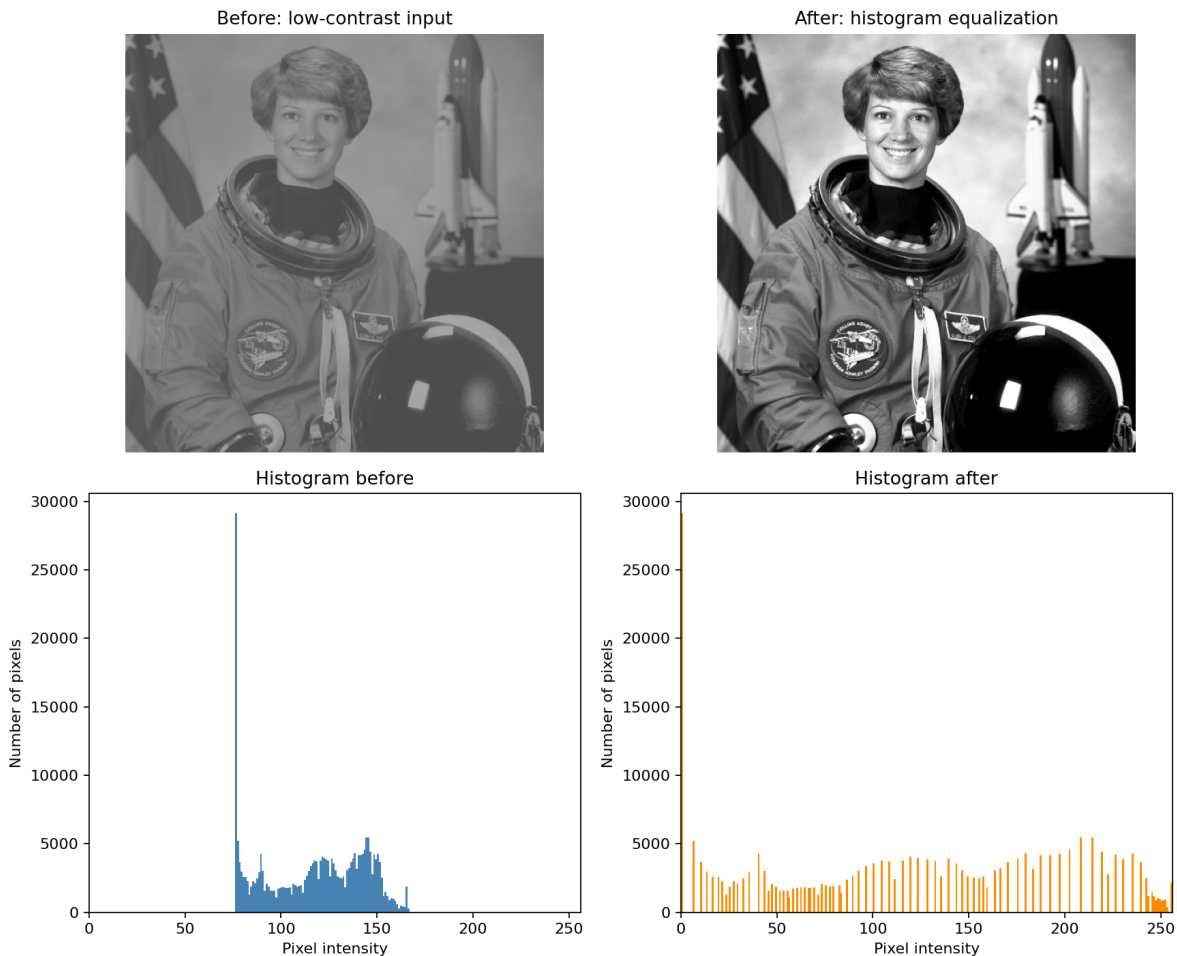


Figure 1. Images and histograms before and after enhancement. Matching display and histogram scales are used.

Reading the histograms. The horizontal axis shows pixel brightness: 0 is black and 255 is white. The vertical axis counts pixels. A taller bar means more pixels at that brightness.

Observation. Before enhancement, the image looks gray and faded, with brightness values between 76 and 166. After enhancement, the values extend from 0 to 255. The dark helmet and clothing folds stand out more clearly against the face and background. The wider occupied histogram range agrees with the stronger visible contrast.

Conclusion and limitation. Histogram equalization improves contrast for this sample. The histogram still contains spikes and gaps; equal bar heights are not guaranteed. Some regions become very dark, so stronger contrast may look less natural. The method does not create missing detail, and the image should be judged visually as well as through its histogram.

Source: NASA photograph of Eileen Collins, distributed as the public-domain scikit-image astronaut sample.
<https://scikit-image.org/docs/stable/api/skimage.data.html#skimage.data.astronaut>